

Reaz Elias

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Data Scientist with over 4 years of hands-on experience delivering impactful business solutions at a Toronto startup. Expertise spans the complete data science lifecycle, including data wrangling, visualization, advanced modeling (Random Forest, XGBoost, ANN), deployment, and continuous monitoring. Proficient in Python, SQL, Applied Statistics, Machine Learning, and MLOps Cloud services.

PROFESSIONAL EXPERIENCE

Government-affiliated Fisheries Department (via internal referral), ON

05/2024 – present

Freelance Data Scientist

- Spatio-temporal modeling in INLA: Collaborated on spatio-temporal modeling in R using INLA to assess phosphorus impact on fish biomass. Supported a fisheries analyst in building Bayesian spatial models, leveraging Generative AI (e.g., ChatGPT) and documentation to guide design, interpret outputs, and present insights
- Unified Fish Database Development: Designed and built a unified fish database for Fisheries and Oceans Canada (DFO), integrating data from multiple US and Canada sources for Lake Erie. Tools used: Python, pandas, SQLite.

Pitstop Connect, ON, Toronto

09/2020 – 01/2024

Data Scientist

- Predictive maintenance model: Reduced unexpected vehicle breakdowns by 50% through early failure detection by identifying patterns in diagnostic fault codes (DTC) to surface potential subsystem failures early.
 - Alert for engine starting issues: Reduced winter-related failures by 40%. Developed a predictive model for parked vehicle battery failures, reducing unexpected starting issues.
 - TF-IDF-based text processing model: Parsing diagnostic trouble/fault codes descriptions to infer VMRS (Vehicle Maintenance Reporting Standards) code for improving system fault analysis.
 - Data Analysis: Clean, process, insert fleet service records into the database; Analyze relevancy between Service records & fault codes, prepare relevance matrix; Quality assessment of algorithm alerts; Retrieved historical fault code data via APIs from telematics platforms.
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PROJECTS

Data Science ([Project Portfolio](#))

- Ferry Ticket Forecasting - Toronto Islands: Forecasted daily ticket demand using Prophet and XGBoost, capturing seasonality and holidays. Improved accuracy supported planning and staffing.
 - MIT-IDSS Hackathon - Bullet Train Passengers' Satisfaction: Ranked 10/40 teams with 95.15% accuracy. Used XGBoost to identify top satisfaction drivers: comfort, entertainment, check-in, and cleanliness.
 - Customer Classification - Marketing Campaign: Used Random Forest to predict sign-ups, enabling targeted outreach, reduced mailing costs, and improved ROI.
 - Customer Segmentation - K-Means Clustering: Grouped customers by grocery behavior for personalized marketing.
 - Loyalty Score Prediction: Predicted missing scores using Random Forest, aiding in high-value customer targeting.
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SKILL

- Programming: Python (Base, Anaconda env, Pandas, Scikit-learn, Statsmodels, SciPy, SQLAlchemy, Psycopg2), SQL. Familiar with R. Machine Learning: Random Forest, XGBoost, K-Means clustering, PCA, ANN. Model deployment & Cloud: GitHub Actions, FastAPI, Docker, Streamlit, AWS S3, EC2, Lambda, SageMaker. Statistics & Optimization: A/B Testing, Hypothesis Testing, Bayesian Inference, Time Series Analysis.
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EDUCATION/ CERTIFICATIONS

- PhD (Mechanical & Industrial Engineering) Toronto Metropolitan University (TMU), formerly Ryerson University, Toronto, ON.
- Machine Learning Specialization | [Deeplearning.ai](#)